

Limiting Cases (Part D)

Here are some tests to conclude that the EoM and code are running properly based on intuition. The cases are as follows:

- 1) Zero Friction
- 2) Friction with no initial rotation
- 3) Set CoM on skate

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Case 1: Zero Friction

```
p.Ff = 0;
p.d1 = 0.5;
p.d2 = 0.5;
p.L = p.d1+p.d2;

xc_0 = 0;
thetadot_0 = 0;

inits = [xc_0 yc_0 theta_0 vc_0 thetadot_0];

[t,z] = ode45(@HW27_RHS,[0,sim_time],inits,opts,p);

xcs = z(:,1);
yys = z(:,2);
ths = z(:,3);
vcs = z(:,4);
ws = z(:,5);

xgs = xcs + p.d1*cos(ths);
ygs = ycs + p.d1*sin(ths);
xps = xcs + p.L*cos(ths);
yps = ycs + p.L*sin(ths);

figure
plot(xcs,yys)
hold on
plot(xgs,ygs)
plot(xps,yps)
hold off
legend('Skate','CoM','Locked Wheels')
axis equal
title('Position of Car No Friction')
xlabel('X Pos')
ylabel('Y Pos')
grid on

figure
```

```

plot(t,vcs)
grid on
title('Velocity of Car at Skate No Friction')
xlabel('Time (s)')
ylabel('Velocity (m/s)')

% with some initial omega
thetadot_0 = 0.5;

inits = [xc_0 yc_0 theta_0 vc_0 thetadot_0];

[t,z] = ode45(@HW27_RHS,[0,sim_time],inits,opts,p);

xcs = z(:,1);
ygs = z(:,2);
ths = z(:,3);
vcs = z(:,4);
ws = z(:,5);

xgs = xcs + p.d1*cos(ths);
ygs = ycs + p.d1*sin(ths);
xps = xcs + p.L*cos(ths);
yps = ycs + p.L*sin(ths);

figure
plot(xcs,ycs)
hold on
plot(xgs,ygs)
plot(xps,yps)
hold off
legend('Skate','CoM','Locked Wheels')
axis equal
title('Position of Car No Friction')
xlabel('X Pos')
ylabel('Y Pos')
grid on

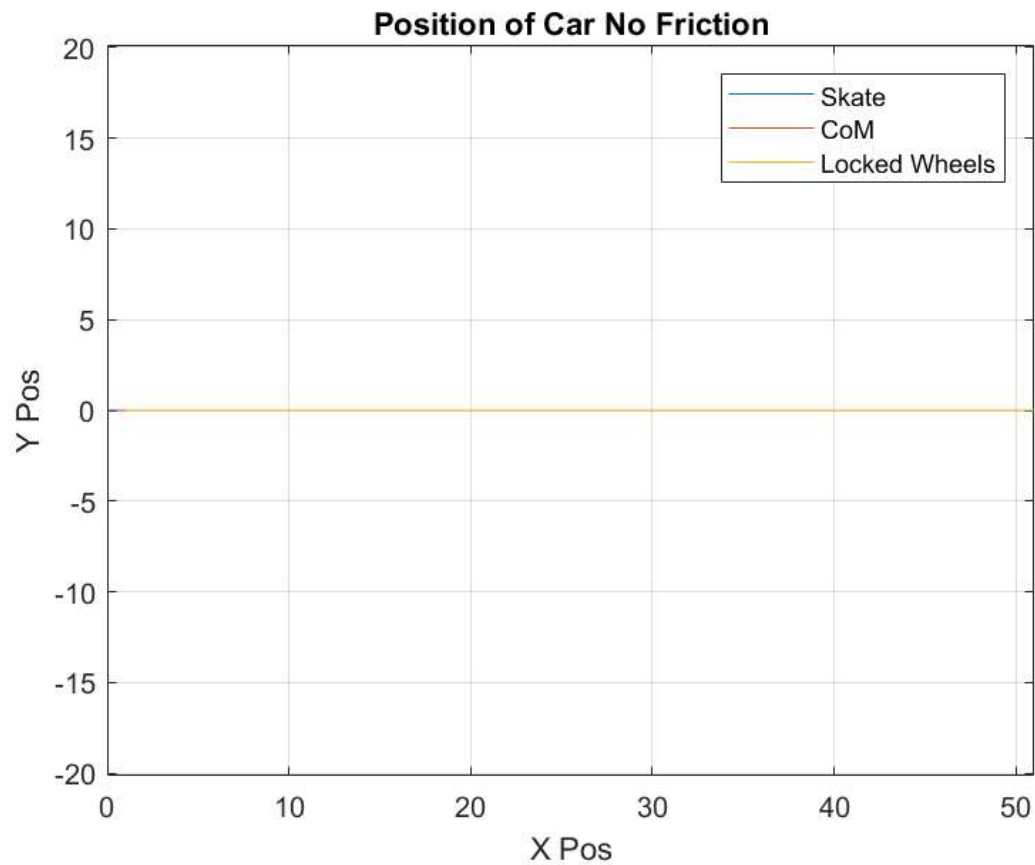
figure
plot(t,vcs)
grid on
title('Velocity of Car at Skate No Friction')
xlabel('Time (s)')
ylabel('Velocity (m/s)')

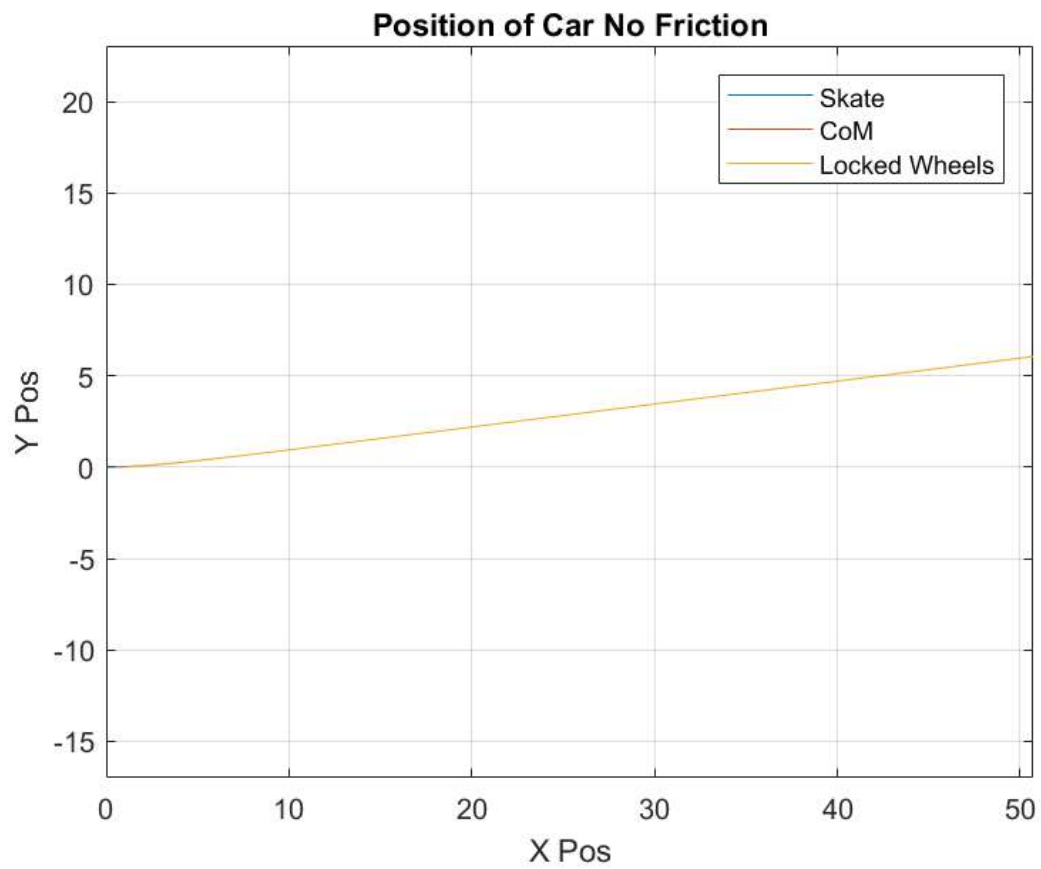
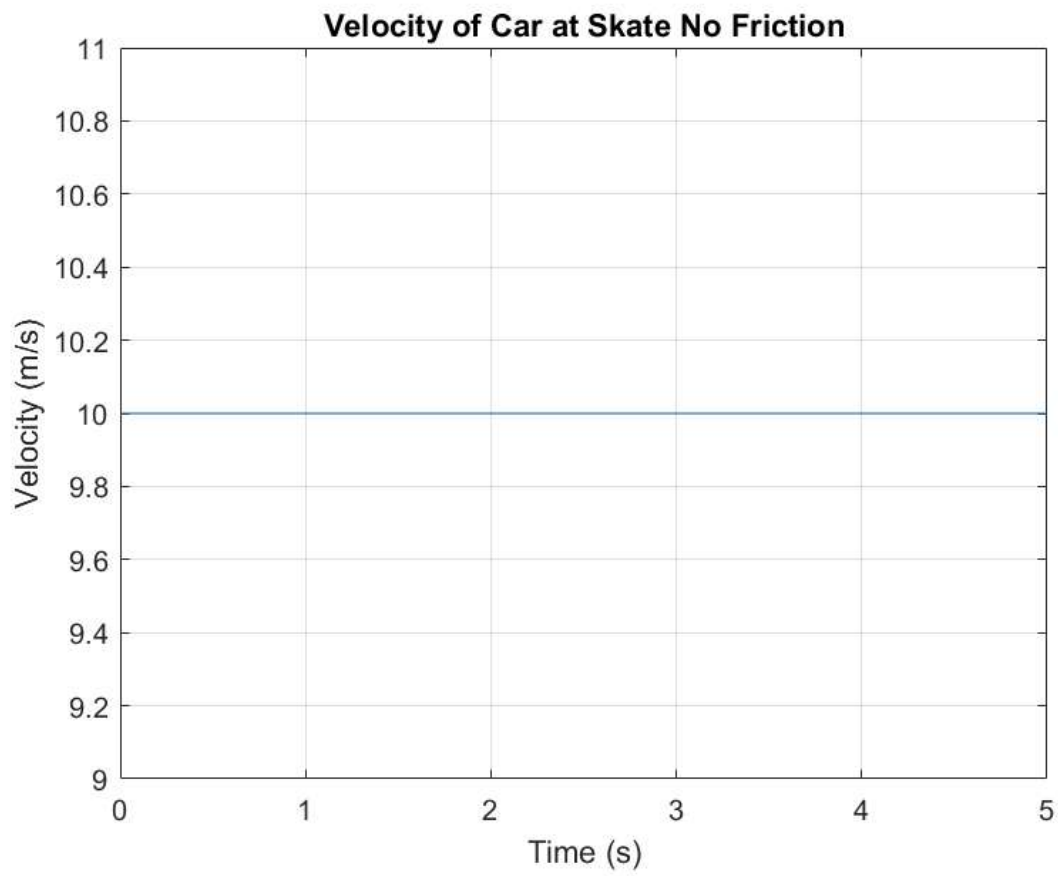
figure
subplot(1,2,1)
plot(t,ths)
title('Pointing Angle No Friction')
xlabel('Time (s)')
ylabel('Angle (rad)')
axis equal
grid on
subplot(1,2,2)
plot(t,ws)
title('Angular Velocity No Friction')
xlabel('Time (s)')
ylabel('Angular Velocity (rad/s)')

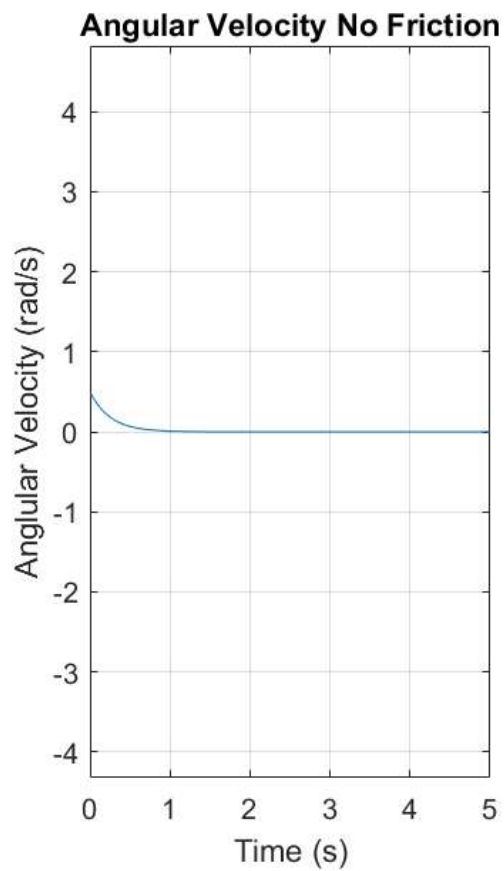
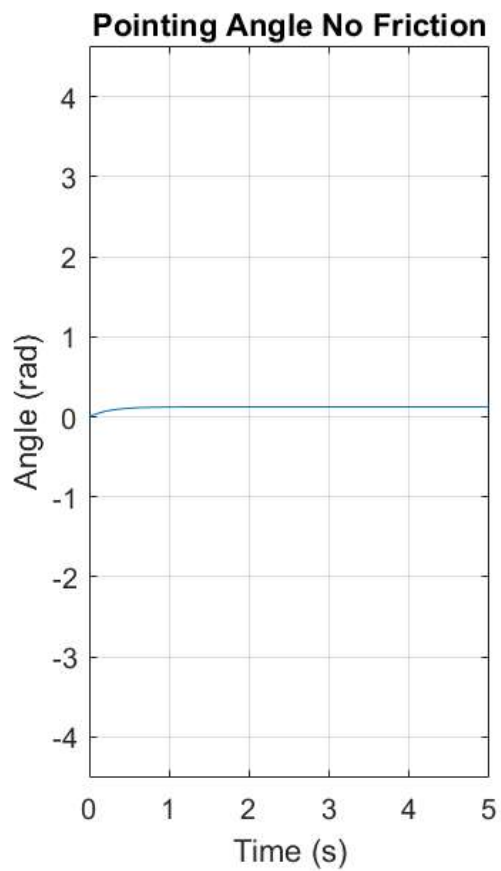
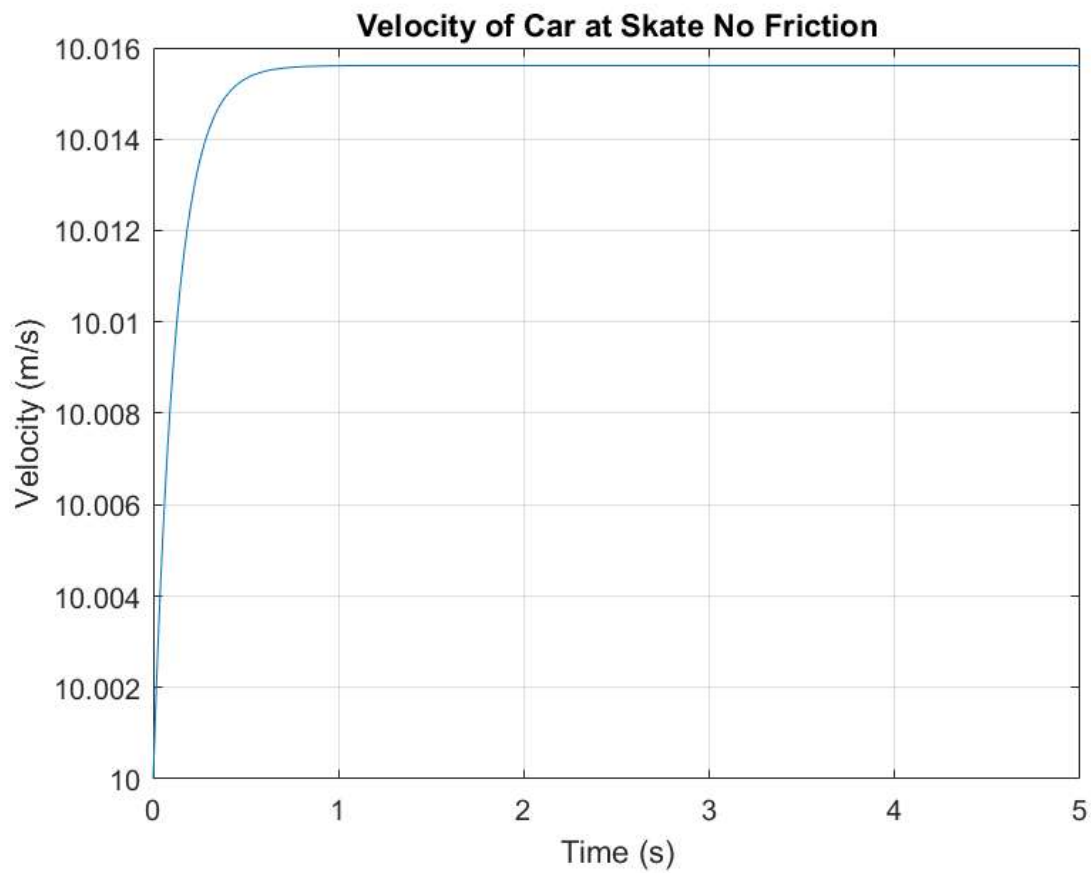
```

```
axis equal  
grid on
```

```
% With no friction, the velocity of the car maintains it's initial given  
% value as well as maintains a straight path through time. Even with some  
% initial angular velocity, the car does not slow down, rather the angular  
% velocity goes to zero but slightly increases the car velocity. Energy is  
% conserved in this case.
```







Case 2: Friction with no initial rotation

```

p.Ff = mu*p.m*p.g;
thetadot_0 = 0;

inits = [xc_0 yc_0 theta_0 vc_0 thetadot_0];

[t,z] = ode45(@HW27_RHS,[0,sim_time],inits,opts,p);

xcs = z(:,1);
ygs = z(:,2);
ths = z(:,3);
vcs = z(:,4);
ws = z(:,5);

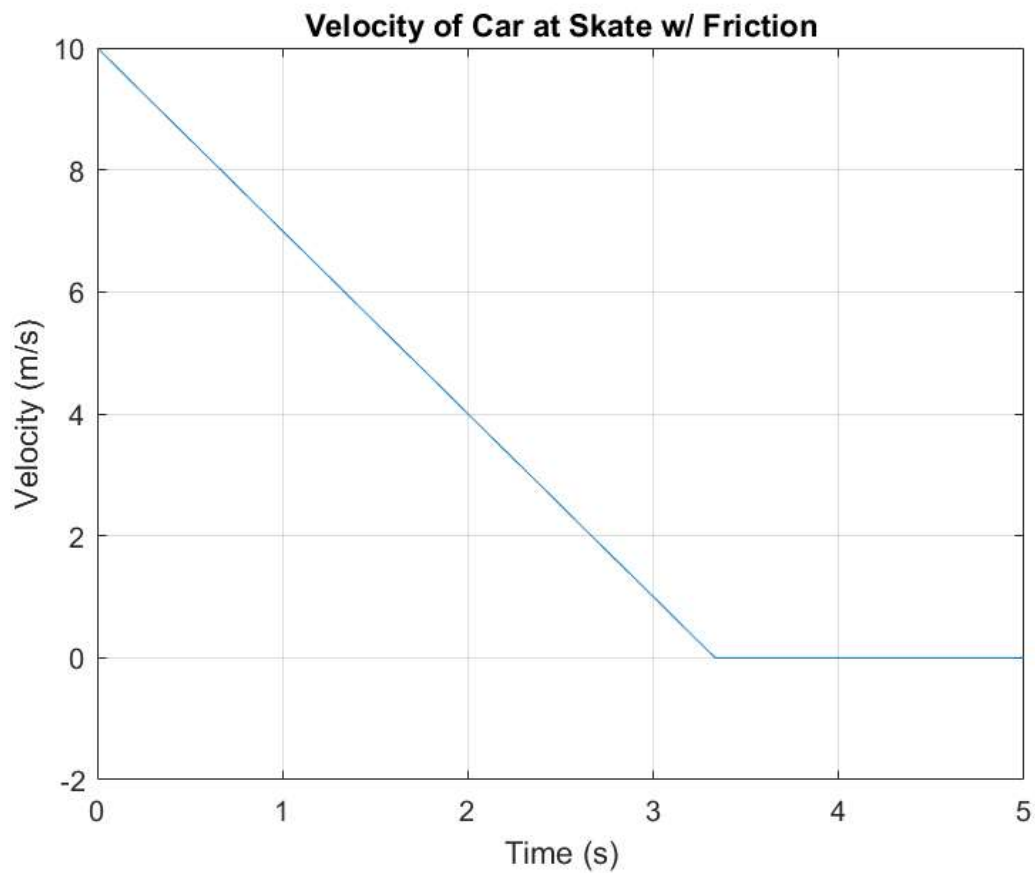
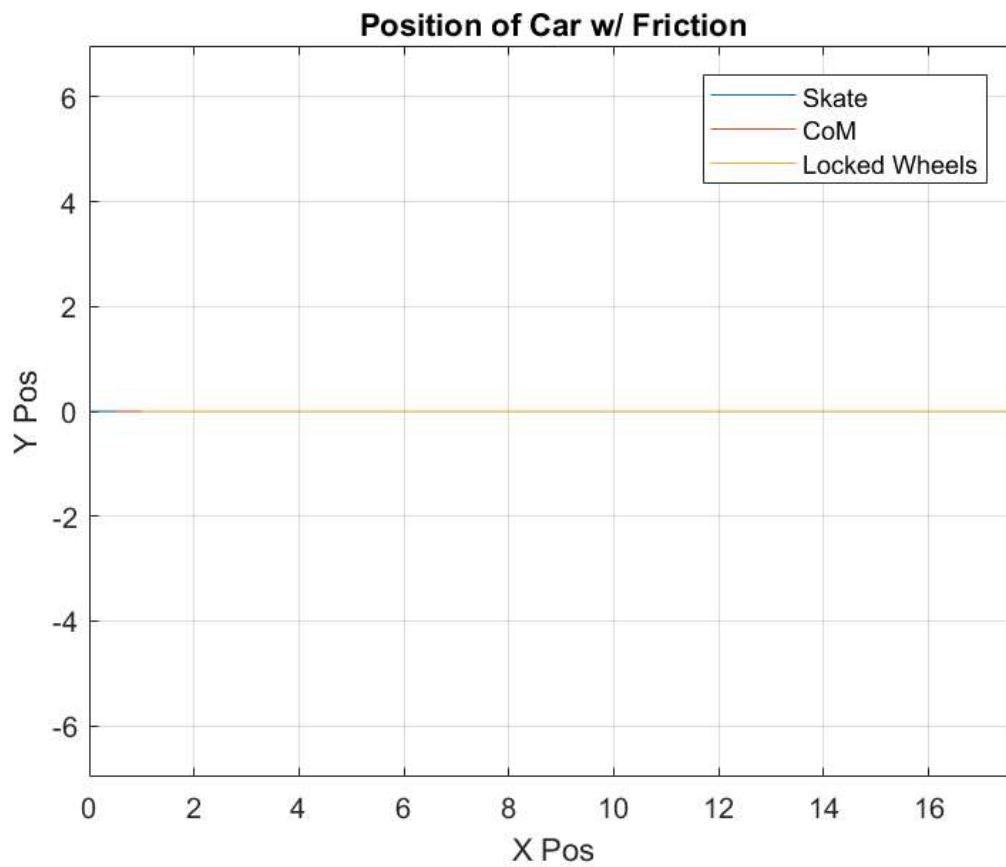
xgs = xcs + p.dl*cos(ths);
ygs = ycs + p.dl*sin(ths);
xps = xcs + p.L*cos(ths);
yps = ycs + p.L*sin(ths);

figure
plot(xcs,ygs)
hold on
plot(xgs,ygs)
plot(xps,yps)
hold off
legend('Skate','CoM','Locked Wheels')
axis equal
title('Position of Car w/ Friction')
xlabel('X Pos')
ylabel('Y Pos')
grid on

figure
plot(t,vcs)
grid on
title('Velocity of Car at Skate w/ Friction')
xlabel('Time (s)')
ylabel('Velocity (m/s)')

% The car's velocity decreases linearly with time or at constant
% deacceleration.

```



Case 3: Set CoM on Skate no Friction

```

p.d1 = 0;
p.d2 = -1;
p.L = p.d1+p.d2;

xc_0 = 1;
thetadot_0 = 0.5;

inits = [xc_0 yc_0 theta_0 vc_0 thetadot_0];

[t,z] = ode45(@HW27_RHS,[0,sim_time],inits,opts,p);

xcs = z(:,1);
ygs = z(:,2);
ths = z(:,3);
vcs = z(:,4);
ws = z(:,5);

xgs = xcs + p.d1*cos(ths);
ygs = ycs + p.d1*sin(ths);
xps = xcs + p.L*cos(ths);
yps = ycs + p.L*sin(ths);

figure
plot(xcs,ygs)
hold on
plot(xgs,ygs)
plot(xps,yps)
hold off
legend('Skate','CoM','Locked Wheels')
axis equal
title('Position of Car')
xlabel('X Pos')
ylabel('Y Pos')
grid on

% here the car follows a stable path despite the skate leading. The problem
% is set up such that there is no weight of the car over the skidding
% wheels thus they effectively drag behind the car with zero effect on the
% system

```

